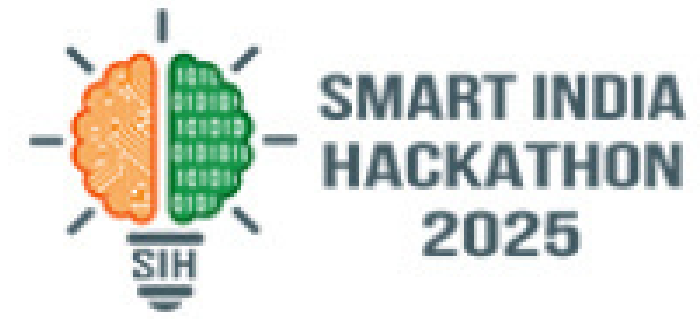


SMART INDIA HACKATHON 2025



- Problem Statement ID - **SIH25124**
- Problem Statement Title - **Student Innovation: Swadeshi for Atmanirbhar Bharat - Toys & Games**
- Theme - **Toys & Games**
- PS Category- **Hardware**
- Team ID - **68037**
- Team Name - **RESISTOR**



MOTION-CONTROLLED INTERACTIVE BRICK BREAKER

DETAILED EXPLANATION OF THE PROPOSED SOLUTION

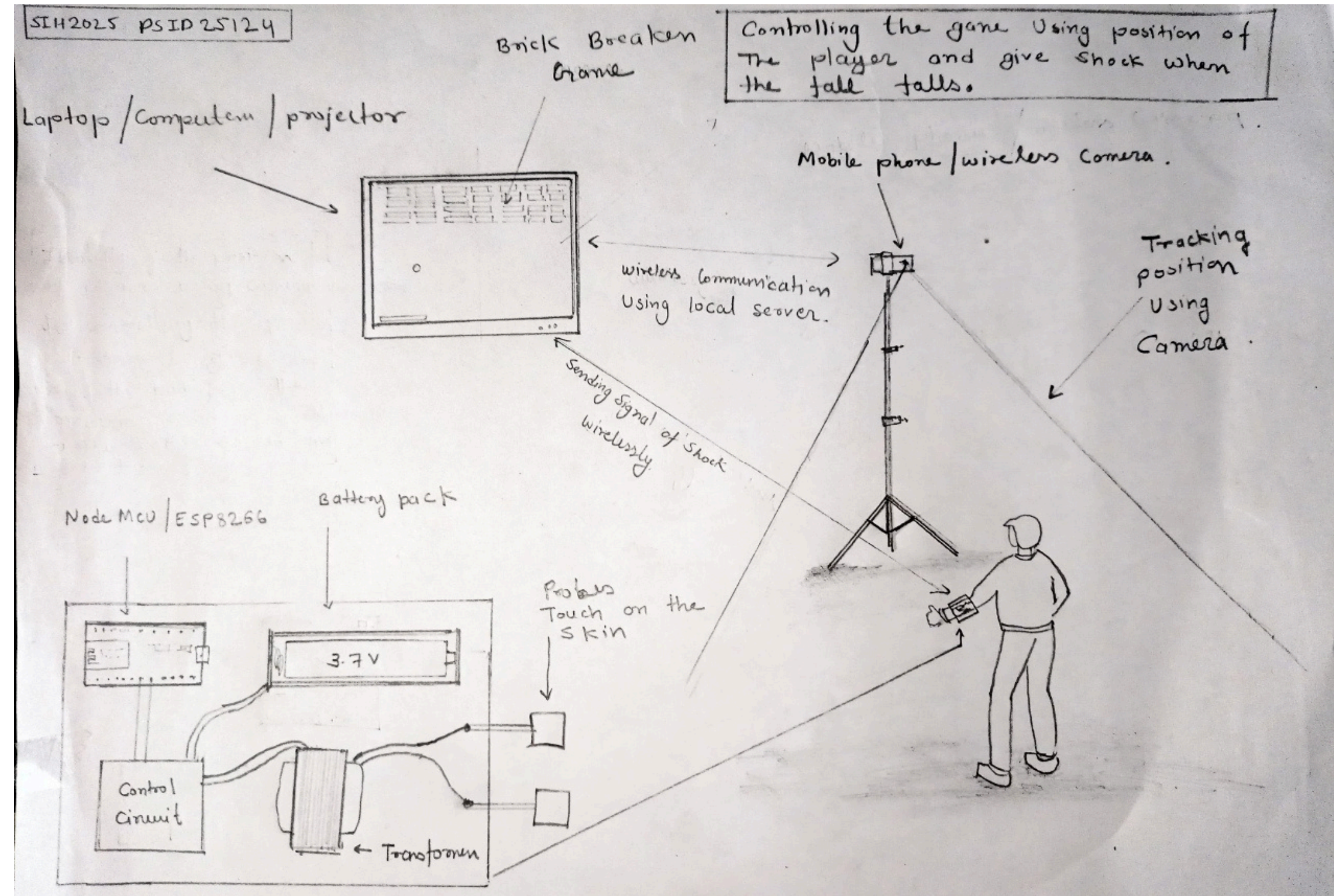
Our project is an interactive version of the classic Brick Breaker game where the player's body movements control the paddle. We use a camera to track the player's horizontal position and map it to the base at the bottom of the screen, so moving left or right moves the paddle in real time. If the ball falls, a wearable device tied to the player's wrist gives a small electric shock. By combining motion tracking with real-world feedback, our game creates a more immersive and engaging experience.

HOW IT ADDRESS THE PROBLEM

Our game stands out because it uses body movement instead of a controller and gives real physical feedback through a shock device. This makes it more immersive, engaging, and challenging compared to regular computer or movie-based games.

INNOVATION AND UNIQUENESS OF THE SOLUTION

The uniqueness of our idea is that it turns the player's body into the controller using camera tracking and adds a wearable shock device for real feedback. This innovative mix makes the game more immersive and engaging than traditional computer games.

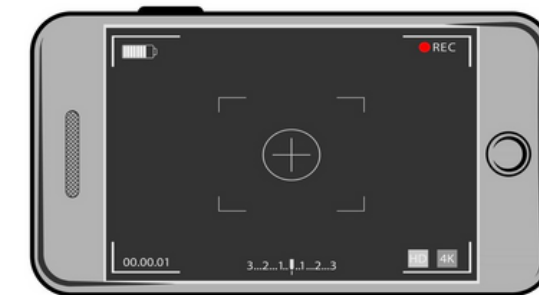


TECHNOLOGIES TO BE USED (E.G. PROGRAMMING LANGUAGES, FRAMEWORKS, HARDWARE)

Our project uses Python to capture and process camera feed for real-time player tracking, using a normal mobile phone as a low-cost camera. A NodeMCU provides wireless communication between the game and hardware. For the shock feedback, a transistor–transformer circuit steps up 1.5V to 100–200V, enabling an effective and affordable interactive system.

METHODOLOGY AND PROCESS FOR IMPLEMENTATION (FLOW CHARTS/IMAGES/WORKING PROTOTYPE)

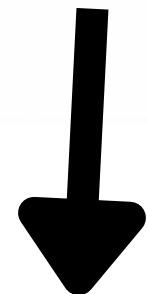
Our methodology involves capturing the player's movement using a mobile phone camera and processing it with Python to track position. This position data is sent wirelessly via NodeMCU to control the paddle in the game. When the ball falls, the NodeMCU triggers a transistor–transformer circuit to deliver a shock through the wearable device. This process creates a real-time, interactive feedback loop between the player and the game.



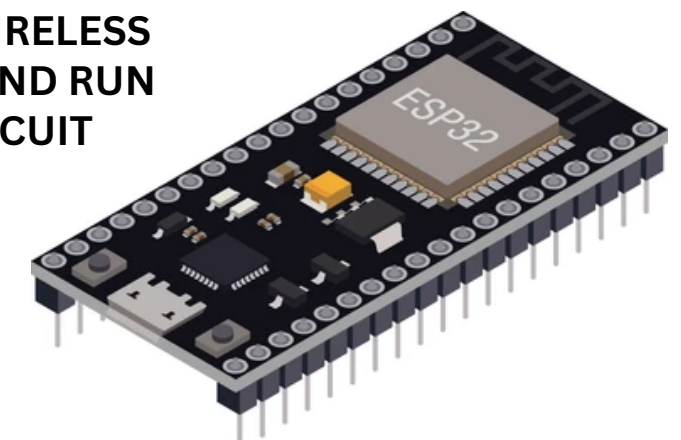
SMARTPHONE FOR VIDEO CAPTURE



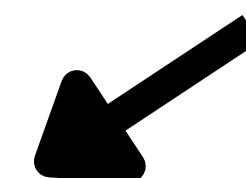
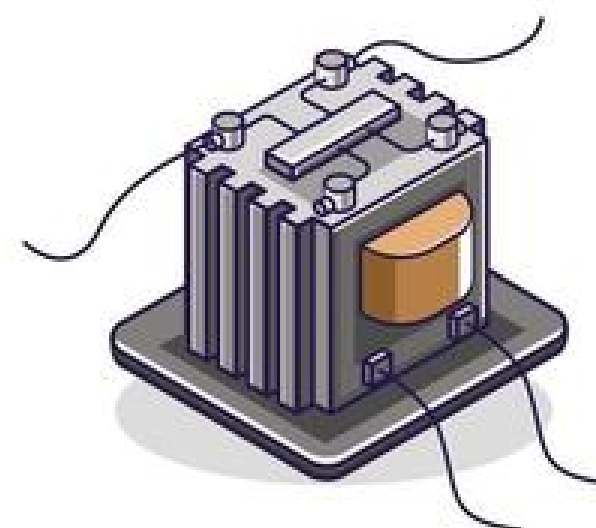
LAPTOP FOR RUN GAME AND PROCESS DATA



NODEMUC FOR WIRELESS COMMUNICATION AND RUN THE SHOCK CIRCUIT



CIRCUIT TO GIVE SAFE ELECTRIC SHOCK SHOCK



ANALYSIS OF THE FEASIBILITY OF THE IDEA

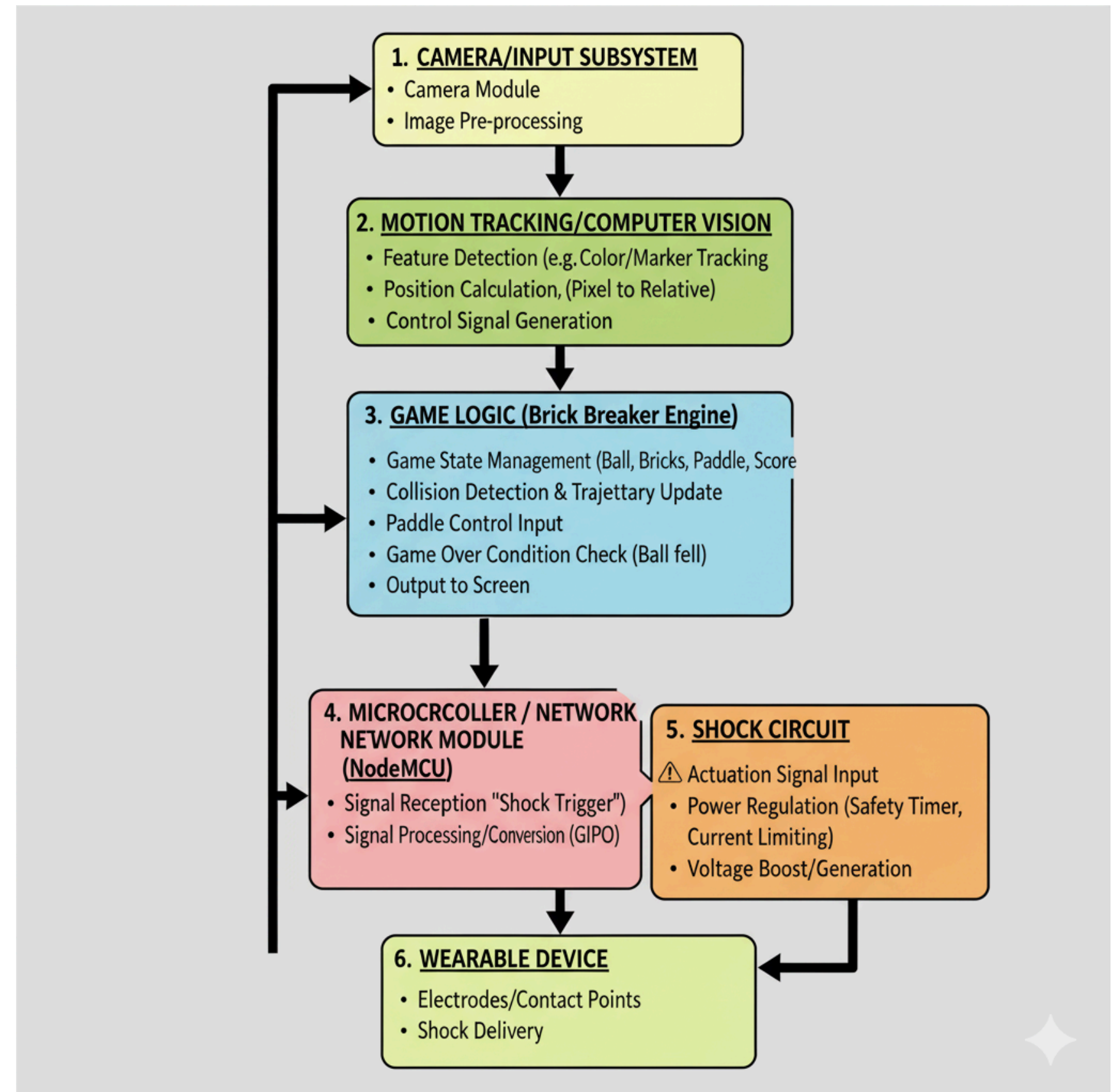
The idea is feasible using low-cost tools like a mobile phone, Python, and NodeMCU. Key challenges include accurate motion tracking, safe shock delivery, and reliable wireless control, but these can be managed with proper design and testing.

POTENTIAL CHALLENGES AND RISKS

Potential challenges include ensuring accurate and fast motion tracking, maintaining safe shock levels to avoid injury, handling wireless communication delays, and designing a reliable shock circuit. Safety, calibration, and real-time performance are key risks that need careful management.

STRATEGIES FOR OVERCOMING THESE CHALLENGES

We can overcome these challenges by using efficient motion-tracking libraries like OpenCV or MediaPipe for accuracy, setting strict safety limits for shock intensity, and thoroughly testing the wearable device. To reduce delays, we can optimize wireless communication with NodeMCU and ensure the shock circuit is well-designed and insulated. Regular calibration and safety checks will ensure reliability and player safety.

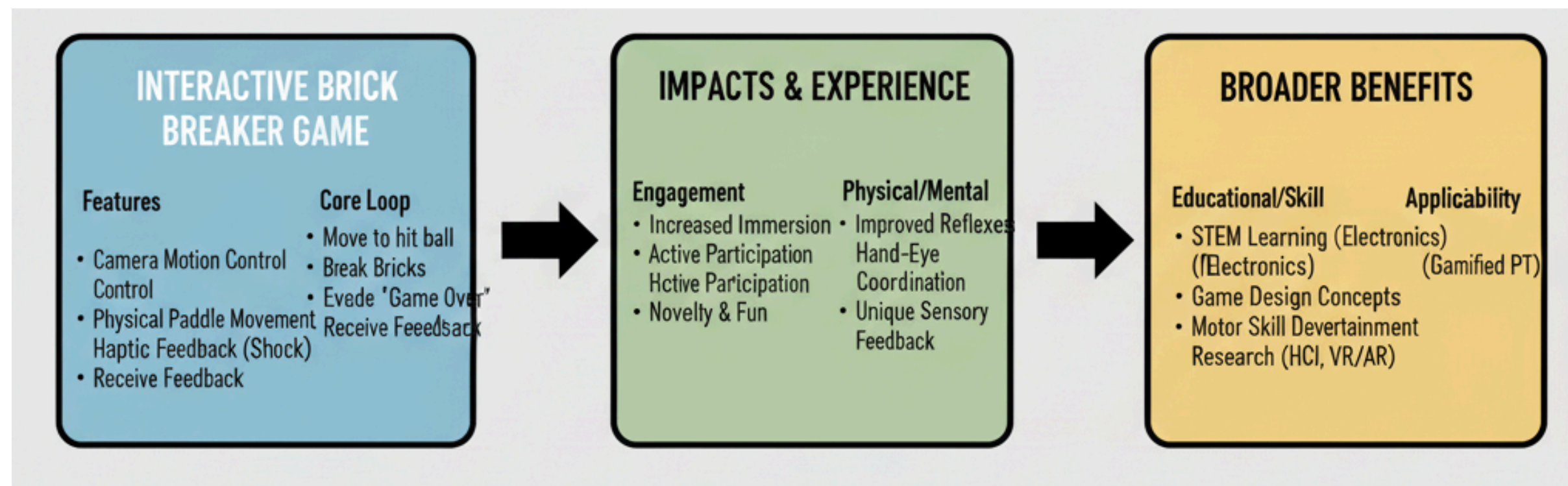


POTENTIAL IMPACT ON THE TARGET AUDIENCE

The game can create a highly engaging and immersive experience for the target audience by combining physical movement with real-time feedback. It can make gaming more interactive, fun, and challenging, appealing especially to players who enjoy active and novel gameplay. However, it may also raise safety concerns, so proper awareness and precautions are important for positive impact.

BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL, ETC.)

- Promotes active and immersive gameplay
- Low-cost and accessible using common devices
- Encourages innovation in motion tracking and wearable feedback
- Reduces need for specialized hardware, lowering environmental impact



RESEARCH AND REFERENCES

OUR RESEARCH PAPER ON THIS GAME

<https://drive.google.com/drive/folders/1G29nt4VKzHTo9oqD-fiQCFsSQS6Me62T?usp=sharing>

THANK YOU